

SI 427 Quiz 1 Solutions
August 23, 2025

Question 1.

1. (2 marks) A coin is tossed repeatedly until two heads appear in a row or two tails appear in a row. Describe a discrete probability space (Ω, p) that models this situation.

Solution:

We will denote by $(01)^k$ the sequence $\underbrace{01 \cdots 01}_{k \text{ times}}$ and by $(10)^k$ the sequence $\underbrace{10 \cdots 10}_{k \text{ times}}$. For $k = 0, 1, 2, \dots$, let

$$\omega_1^k = (01)^k 1, \omega_2^k = (10)^k 0, \omega_3^k = (01)^k 00, \omega_4^k = (01)^k 11.$$

Let $(01)^*$ and $(10)^*$ be the infinite sequences $0101 \dots$ and $1010 \dots$ respectively. We define

$$\Omega := \{(01)^\infty\} \cup \{(10)^\infty\} \cup \left(\bigcup_{k=0}^{\infty} (\{\omega_1^k\} \cup \{\omega_2^k\} \cup \{\omega_3^k\} \cup \{\omega_4^k\}) \right). \quad (1 \text{ mark})$$

We define $p((01)^*) = p((10)^*) = 0$. Now, let p be the probability of heads appearing and $q = 1 - p$ be the probability of tails appearing when the coin is tossed. We define

$$p(\omega_1^k) = p^{k+1} q^k = p(pq)^k, \quad k \geq 1, \quad \sum_{k \geq 1} p(\omega_1^k) = \frac{p^2 q}{1 - pq} = S_1,$$

$$p(\omega_2^k) = p^k q^{k+1} = q(pq)^k, \quad k \geq 1, \quad \sum_{k \geq 1} p(\omega_2^k) = \frac{q^2 p}{1 - pq} = S_2,$$

$$p(\omega_3^k) = p^k q^{k+2} = q^2 (pq)^k, \quad k \geq 0, \quad \sum_{k \geq 0} p(\omega_3^k) = \frac{q^2}{1 - pq} = S_3,$$

$$p(\omega_4^k) = p^{k+2} q^k = p^2 (pq)^k, \quad k \geq 0, \quad \sum_{k \geq 0} p(\omega_4^k) = \frac{p^2}{1 - pq} = S_4.$$

(1 mark)

$$\sum_{\omega} p(\omega) = S_1 + S_2 + S_3 + S_4 = \frac{p^2 q + q^2 p + p^2 + q^2}{1 - pq} = \frac{(p + q)^2 - 2pq + pq(p + q)}{1 - pq} = \frac{1 - pq}{1 - pq} = 1.$$

If $p = q = 1/2$ we have $p(\omega) = \frac{1}{2^n}$, if ω is a sequence of length $n \geq 2$.

Comments:

- (a) Most students have not included $(01)^*$ and $(10)^*$ in the sample space. In this case, I have cut a 1/2 mark (when correcting the papers, I used the notation $(01)^\infty$ for $(01)^*$ and $(10)^\infty$ for $(10)^*$).
- (b) Very few students have treated a biased coin. Many have taken a fair coin. I have not penalised students for this.
- (c) If the elementary probabilities have been correctly specified, I have given full marks. I have not cut marks if students have not verified that $\sum_{\omega} p(\omega) = 1$.

2. (3 marks) Consider a disease X that affects 1 in 10 people. A medical test to determine the presence of the disease is 60% accurate in the sense that if a person does not have the disease, the chance that the test is positive, that is, it asserts the presence of the disease, is 40%, and if the person has the disease, the chance that the test is negative, that is, it asserts that the disease is not present, is 40%. Suppose a person is tested for X and the result is positive. What is the chance that the person has the disease?

Solution: Let A be the event that a person has the disease X . Let B be the event that a person tests positive for X .

We are given that $P(A) = 1/10$ and $P(B | A) = 3/5$ and $P(B | A')P(A) = 3/50$. (1 mark)

Since A and A' are mutually exclusive and exhaustive events,

$$P(B) = P(B | A)P(A) + P(B | A')P(A') = 3/5 \cdot 1/10 + 2/5 \cdot 9/10 = 21/50. \quad (1 \text{ mark})$$

Now, $P(A | B) = \frac{P(B|A)P(A)}{P(B)}$ by Bayes' Rule. Hence,

$$P(A | B) = \frac{3/50}{21/50} = 1/7. \quad (1 \text{ mark})$$

Comments: Sometimes I have awarded only 1/2 a mark if there were errors in some of the steps above. The different mistakes students have made and the marks cut in various cases are too numerous to list here.

Question 2.

1. (2 marks) Prove that a countable union of countable sets is countable.

Solution: This question was solved in the tutorial. Let A_m , $m \in \mathbb{N}$ be the given countable collection of sets. Assume that the A_m are (pairwise) disjoint, that is, $A_i \cap A_j = \emptyset$ for all $i, j \in \mathbb{N}, i \neq j$. Since A_m is a countable set, there is a bijective map $n \mapsto a_{mn}$ between $\mathbb{N}$ and the elements of A_m . We define a function $f : \bigsqcup_{m=1}^{\infty} A_m \rightarrow \mathbb{N} \times \mathbb{N}$ by $f(a_{mn}) = (m, n)$. Note that f is well-defined since $a_{m_1 n_1} = a_{m_2 n_2}$ implies $m_1 = m_2$ (since the sets A_m are pairwise disjoint), and $n_1 = n_2$. The function f is clearly injective since $(m_1, n_1) = (m_2, n_2)$ implies that $m_1 = m_2$ and $n_1 = n_2$, so $a_{m_1 n_1} = a_{m_2 n_2}$. Since $\mathbb{N} \times \mathbb{N}$ is countable, $\bigsqcup_{m=1}^{\infty} A_m$ is countable. (1 mark)

If the given sets A_m are not disjoint, we define $B_1 = A_1$ and $B_m = A_m \setminus \bigcup_{k=1}^{m-1} A_k$. Then, the sets B_m are pairwise disjoint, and $\bigcup_{m=1}^{\infty} B_m = \bigcup_{m=1}^{\infty} A_m$. (1 mark)

2. (1 mark) Let (Ω, p) be a discrete probability space and let A and B be events in Ω . If the occurrence of B makes A more likely, will the occurrence of A make B more likely?

Comments:

- (a) Only one student gave the first part of the argument, but not the second (and has thus received 1 mark).
- (b) Many students did not assume that the sets A_m were disjoint, but nonetheless defined the function f as above and asserted that it was injective. They have received no marks, since they have failed to realise that the map was not well-defined. Some students gave a map $g : \mathbb{N} \times \mathbb{N} \rightarrow \bigcup_{m=1}^{\infty} A_m$, by defining $g(m, n) = a_{mn}$ and asserting that it is injective. Again, this is not true unless the sets A_m have been assumed to be pairwise disjoint.
- (c) A small subset of students represented the elements of the union as a matrix array and drew arrows as when counting the rational numbers. As long as they did not make an incorrect statement, they were awarded a 1/2 mark, but not more.

Solution: If the occurrence of B makes A more likely, we have $P(A|B) > P(A)$, that is,

$$\frac{P(A \cap B)}{P(B)} > P(A).$$

It follows that

$$\frac{P(A \cap B)}{P(A)} = P(B|A) > P(B),$$

which means that the occurrence of A makes B more likely. (1 mark)

Comment: There is no partial credit for this question. Many students have given the definitions of the two conditional probabilities, but if there is no explicit inequality, no marks have been awarded.

3. (2 marks) With notation as in the previous question, if $P(A) = 0.6$ and $P(B) = 0.8$, can A and B be mutually exclusive? Show that $P(B|A) \geq 2/3$.

Solution:

We have

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) \geq 0.6 + 0.8 - 1 = 0.4,$$

since $P(A \cup B) \leq 1$. Thus, A and B cannot be mutually exclusive, since this would imply $P(A \cup B) = 0$. (1 mark)

Further, $P(B | A) = \frac{P(B \cap A)}{P(A)} \geq 0.4/0.6 = 2/3$. (1 mark)

Solution:

We have

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) \geq 0.6 + 0.8 - 1 = 0.4,$$

since $P(A \cup B) \leq 1$. Thus, A and B cannot be mutually exclusive, since this would imply $P(A \cup B) = 0$. (1 mark)

Further, $P(B | A) = \frac{P(B \cap A)}{P(A)} \geq 0.4/0.6 = 2/3$. (1 mark)

Question 3.

1. (3 marks) Derive a formula for the probability P_m that a permutation chosen randomly from S_m , the group of permutations of $[m]$, is a derangement (that is, a permutation σ such that $\sigma(i) \neq i$ for all $1 \leq i \leq m$).

Solution: Let $A_i = \{\sigma \mid \sigma(i) = i\}$, $1 \leq i \leq n$. If A is the set of derangements of S_m , we see that $A' = \bigcup_{i=1}^m A_i$. Using the inclusion-exclusion principle, we have

$$P(A') = \sum_{k=1}^m (-1)^{k-1} S_k, \quad (1 \text{ mark})$$

where $S_k = \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq m} P(A_{i_1} \cap \dots \cap A_{i_k})$. We have $P(A_{i_1} \cap \dots \cap A_{i_k}) = \frac{(m-k)!}{m!}$. There are $\binom{m}{k}$ subcollections of the A_i containing k of these events. It follows that

$$S_k = \binom{m}{k} \frac{(m-k)!}{m!} = \frac{1}{k!}. \quad (1 \text{ mark})$$

Hence, $P(A') = \sum_{k=1}^m (-1)^{k-1} \frac{1}{k!}$, so

$$P_m = P(A) = 1 - \sum_{k=1}^m (-1)^{k-1} \frac{1}{k!} = \sum_{k=0}^m \frac{(-1)^k}{k!}. \quad (1 \text{ mark})$$

Comments:

- (a) It is important to specify the events A_i correctly before applying the inclusion-exclusion principle. If the event A_i has been incorrectly specified, no marks have been awarded.
 - (b) Sometimes, the event A_i has not been explicitly specified, but the rest of the argument is (more or less) correct. In this case students have been awarded upto 1.5 marks.
2. (2 marks) What is the smallest value of m such that $|P_m - e^{-1}| < 0.01$. Justify.

Solution: Taylor's theorem for the function $f(x) = e^{-x}$ asserts that

$$e^{-x} = \sum_{k=0}^m \frac{(-1)^k}{k!} + R_m(x),$$

where $R_m(x) = \frac{f^{(m+1)}(c)x^{m+1}}{(m+1)!}$ for some $0 < c < x$.

In our case, $x = 1$ and $f^{(m+1)}(c) = (-1)^{m+1}e^{-c}$. (1/2 mark)

Since $e^{-c} < 1$, we have $|R_m(1)| \leq \frac{1}{(m+1)!}$. Hence, $|R_m(1)| < 1/5! < 0.01$ if $m \geq 4$. (1 mark)

If $m < 4$, $|R_m(x)| \geq \frac{e^{-1}}{(m+1)!} > 1/3[(m+1)!] > 1/3 \cdot 24 = 1/72 > 0.01$ (since e^{-c} takes its minimum at $c = 1$). (1/2 mark)

Comments:

- (a) Very few students have got this question correct.
- (b) No marks have been given if the correct formula for the remainder term has not been given. If the correct formula has been given a 1/2 mark has been awarded.
- (c) The question asked for the smallest value of m , so students have received the second mark only if they have verified that $m \leq 3$ does not work.